Tanjim's Professional Knowledge Base

Sample multimodal test document — text, tables, charts & diagrams

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This document is a **sample test file**, generated to help develop and validate the multimodal extraction pipeline for the Basic_Rag_System project. It intentionally mixes content types that a multimodal RAG needs to handle: plain paragraphs, bullet lists, a structured table, and two embedded raster images (a bar chart and an architecture diagram). Replace the placeholder text below with real career details once the pipeline is confirmed to extract and embed each modality correctly.

Professional Summary

Developer focused on building practical, end-to-end software systems, with particular interest in AI/ML pipelines and personal finance tools. Approaches new topics by building real, functional systems rather than isolated exercises — for example, building a hybrid retrieval-augmented generation (RAG) system from scratch before extending it toward multimodal capability.

Core Focus Areas

- Retrieval-Augmented Generation (RAG) systems — hybrid vector + keyword search
- AI/ML pipeline design and deployment (including free-tier hosting)
- Personal finance and stock-tracking tools
- iOS/SwiftUI apps integrating on-device ML (MLX)

Project Experience (sample table)

Tables are a common multimodal challenge — extraction must preserve row/column structure so the embedding model (or a table-aware parser) can represent it meaningfully instead of collapsing it into unstructured text.

Project	Stack	Status	Focus
Basic_Rag_System	Python, LangChain, ChromaDB, BM25	Active	Hybrid vector + keyword retrieval
Multimodal RAG	CLIP, FAISS, Groq API	In progress	PDF text + image extraction
Stock Tracker	HTML/JS, self-contained	Maintained	Personal market tracking
MLX SwiftUI Chat	Swift, MLX, App Intents	Exploring	On-device chat + Siri search

Skill Proficiency (sample chart)

This is a raster image (PNG) embedded in the PDF — useful for testing whether your pipeline correctly detects, extracts, and embeds non-text visual content (e.g. via CLIP or a vision-language captioning step) rather than skipping it.

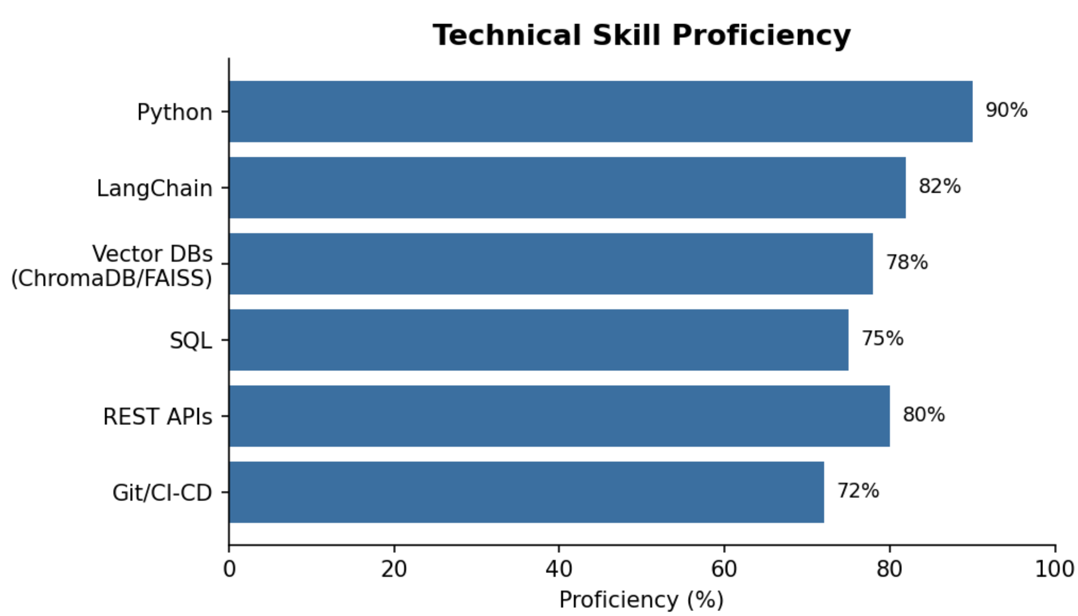

Figure 1 — Sample technical skill proficiency chart.

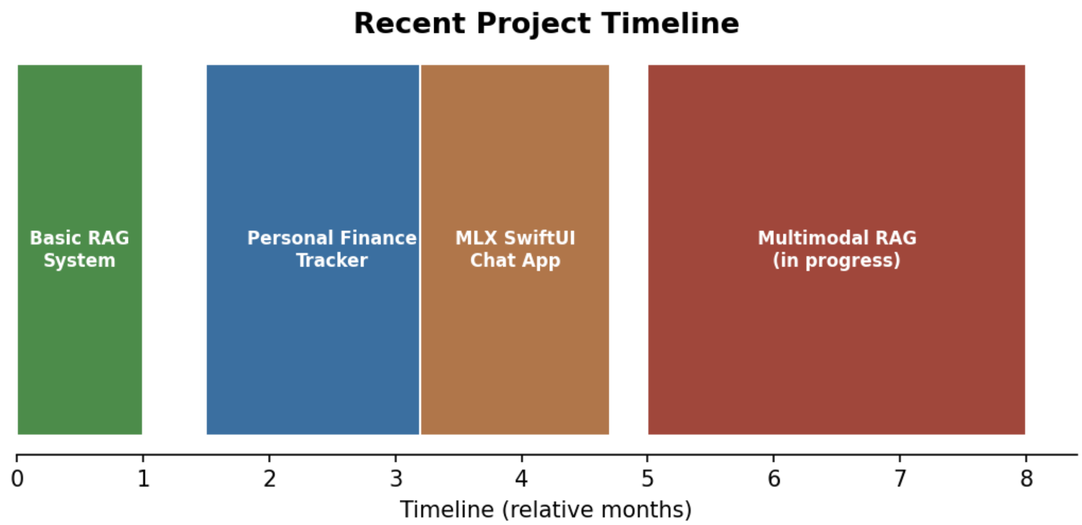

Figure 2 — Sample recent project timeline.

Target Multimodal RAG Architecture (diagram)

A diagram like this tests image-based reasoning: can the multimodal pipeline describe the flow shown in the picture (e.g. via a vision-capable embedding or captioning model) when asked a question about it?

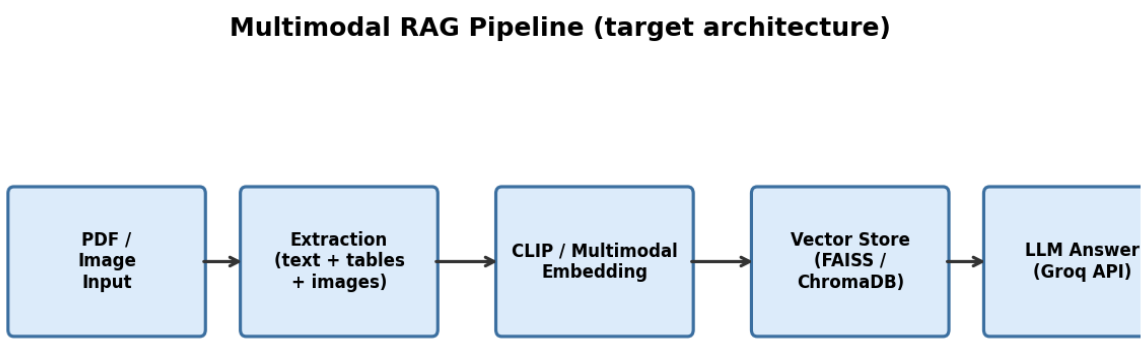

Figure 3 — Target ingestion-to-answer pipeline for the multimodal RAG extension.

How to Use This File

- Run it through your PDF extraction step and confirm text, table rows, and both images are each captured as separate chunks/nodes.
- Embed the text chunks with your text embedder and the two images with a CLIP (or similar) image embedder.
- Store both embedding types in your vector store with a shared metadata schema (e.g. source_page, modality, content_type).
- Ask a question that can only be answered from a chart (e.g. "What is the proficiency percentage for LangChain?") to confirm image-grounded retrieval works end to end.
- Once extraction and retrieval are verified, swap this placeholder content for your real career/professional documents.